

Q5

Find the value of k , if,

i. The roots of the eqn on $5x^2 - 7x + k - 2 = 0$, satisfy the relation $2\alpha + 5\beta = 1$

sol

As per given statement,

$$2\alpha + 5\beta = 1 \quad \text{--- (i)}$$

so

$$5x^2 - 7x + k - 2 = 0$$

$$a = 5$$

$$b = -7$$

$$c = k - 2$$

$$\text{we know that } \alpha + \beta = -\frac{b}{a}$$

$$\alpha + \beta = -\frac{(-7)}{5} = \frac{7}{5}$$

$$\alpha\beta = \frac{c}{a} = \frac{k-2}{5} = \frac{k-2}{5} \quad \text{--- (ii)}$$

→ so we don't have any formula that will make $2\alpha + 5\beta = 1$, therefore we will find the value of either α or β

so,

$$\alpha + \beta = \frac{7}{5}$$

$$\left(\beta = \frac{7}{5} - \alpha \right) \quad \text{--- (ii)}$$

Now put the eqⁱ in eq no (i)

so

$$2\alpha + 5\beta = 1$$

$$2\alpha + 5\left(\frac{7}{5} - \alpha\right) = 1$$

$$2\alpha + \frac{35}{5} - 5\alpha = 1$$

$$2\alpha + 7 - 5\alpha = 1$$

$$-3\alpha = 1 - 7$$

$$-3\alpha = -6$$

$$\boxed{\alpha = 2}$$

Now for finding the value of β put the value in eq no ii

$$\alpha\beta = \frac{7}{5} - 2$$

$$\beta = \frac{7-10}{5}$$

$$\boxed{\beta = -\frac{3}{5}}$$

Now we have to find out the value of k also, by using eq (iii) to find the value of k .

$$\alpha\beta = \frac{k-2}{5} \rightarrow \text{we already find out value of } \alpha\beta.$$

$$\therefore \left(2\right) \left(-\frac{3}{5}\right) = \frac{k-2}{5}$$

$$-\frac{6}{5} = \frac{k-2}{5} \Rightarrow -30 = 5k - 10$$

$$\boxed{5k = -20}$$
$$\boxed{k = -4}$$

Now let's satisfy it

$$2\alpha + 5\beta = 1$$

$$2(2) + 5\left(\frac{-3}{5}\right) = 1$$

$$4 + -3 = 1$$

$$\boxed{1 = 1}$$

Hence satisfied